

Centro de Procesamiento de Datos



**UNIVERSIDAD
DE GRANADA**

Práctica 7. Copias de seguridad
Juan Navarro Maldonado

Introducción.....	2
Vagrantfile.....	2
1.- Fail2ban.....	3
Instalación de fail2ban en Ubuntu.....	3
Comprobación del acceso SSH.....	6
2.- Instalación de Google Authenticator.....	10
Instalación.....	10
3.- (Opcional) Supervisión de NGINX mediante Fail2ban.....	17
Modificación del fichero jail.conf.....	17
Instalación de NGINX en la máquina Ubuntu.....	17
Creación de una página con acceso mediante autenticación.....	18
Comprobación entre nodos el bloqueo en los accesos a las páginas.....	20
4.- (Opcional) Utilizar la red TOR para acceder directamente a la máquina virtual.....	25
Instalación TOR.....	25
Configuración del cliente.....	26
Para configurar el cliente debemos de editar el fichero /etc/tor/torrc y debemos de añadir las siguientes líneas al fichero:.....	26

Introducción

Vagrantfile

La máquina virtual que voy a usar es un ubuntu 18.04 con el siguiente vagrantfile:

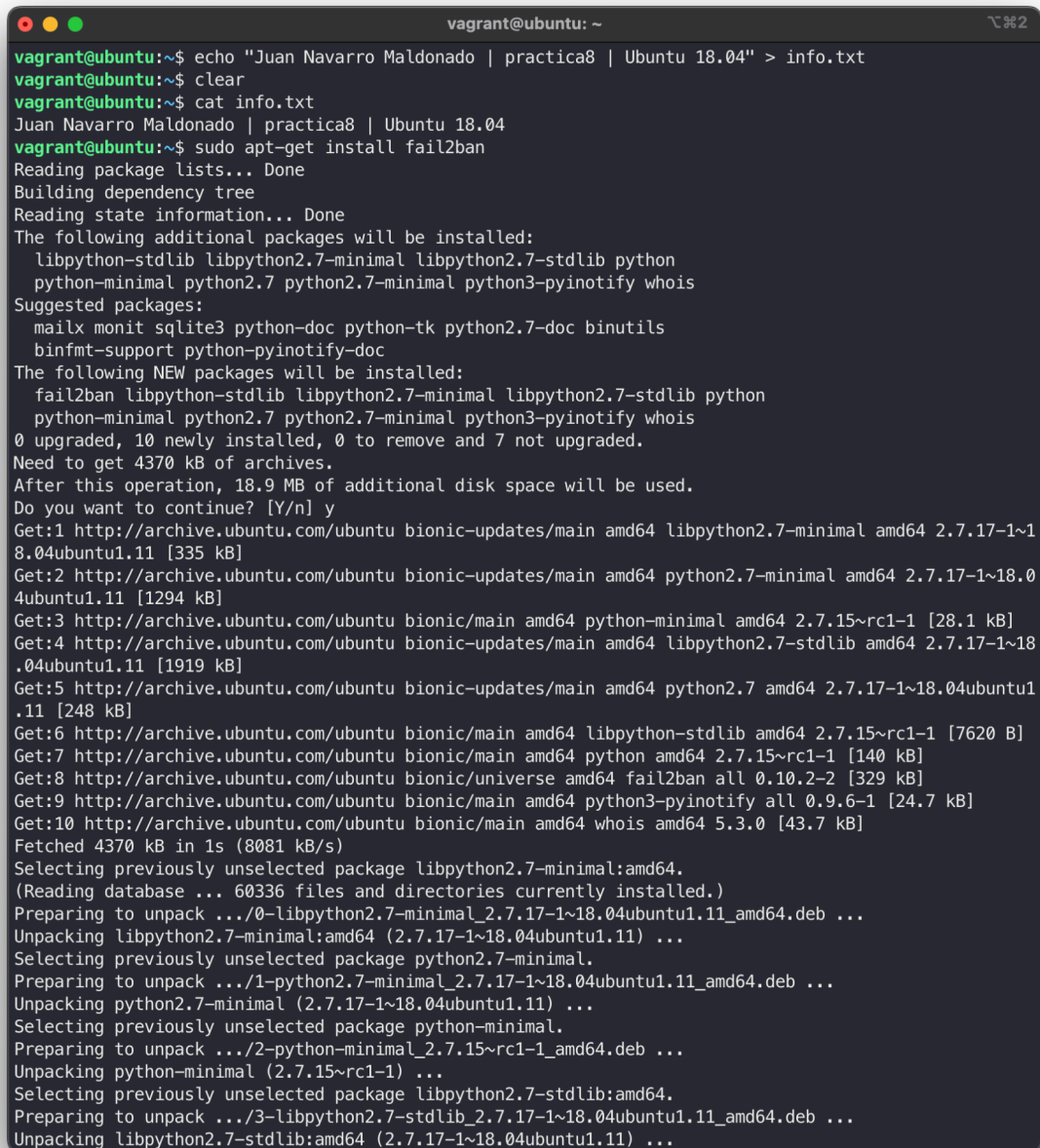
```
1. Vagrant.configure("2") do |config|
2.   config.vm.box = "ubuntu/bionic64" # Ubuntu 18.04
3.   config.vm.network "private_network", type: "dhcp"
4.
5.   config.vm.hostname = "ubuntu"
6.   config.vm.provision "shell", inline: <<-SHELL
7.     sudo apt-get update -y
8.     sudo apt-get upgrade -y
9.   SHELL
10.
11.   config.vm.network "forwarded_port", guest: 80, host: 8080
12.   config.vm.network "forwarded_port", guest: 443, host:
13.     8443
14.   config.vm.provider "virtualbox" do |vb|
15.     vb.memory = "2048"
16.     vb.cpus = 4
17.   end
18. end
```

1.- Fail2ban

Instalación de fail2ban en Ubuntu

Instalaremos Fail2Ban con el siguiente comando:

```
sudo apt-get install fail2ban
```

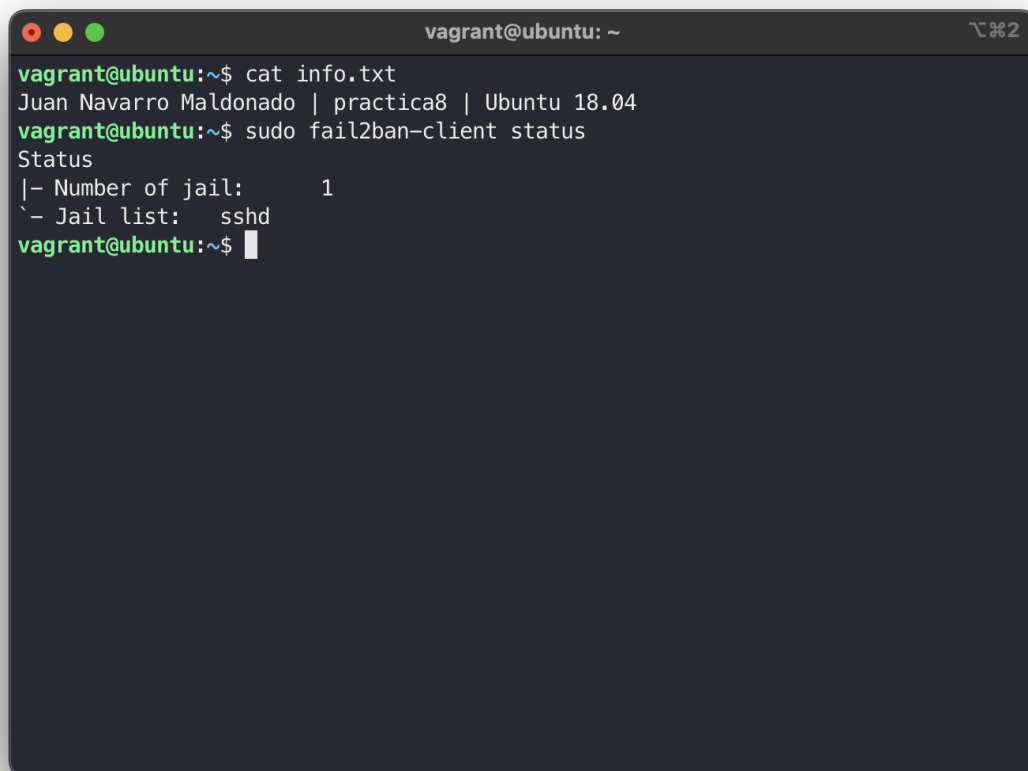
A terminal window titled 'vagrant@ubuntu: ~' showing the installation of fail2ban. The user runs 'echo "Juan Navarro Maldonado | practica8 | Ubuntu 18.04" > info.txt', 'clear', and 'cat info.txt'. Then they run 'sudo apt-get install fail2ban'. The terminal output shows the package lists being read, the dependency tree being built, and the state information being read. It lists additional packages to be installed (libpython-stdlib, libpython2.7-minimal, libpython2.7-stdlib, python, python-minimal, python2.7, python2.7-minimal, python3-pyinotify, whois) and suggested packages (mailx, monit, sqlite3, python-doc, python-tk, python2.7-doc, binutils, binfmt-support, python-pyinotify-doc). It also lists the NEW packages to be installed (fail2ban, libpython-stdlib, libpython2.7-minimal, libpython2.7-stdlib, python, python-minimal, python2.7, python2.7-minimal, python3-pyinotify, whois). The output indicates that 0 packages will be upgraded, 10 will be newly installed, 0 will be removed, and 7 will not be upgraded. The total size of the archives to be installed is 4370 kB, and an additional 18.9 MB of disk space will be used. The user is prompted to continue (Y/n) and they press 'y'. The terminal then shows the progress of downloading the packages from the Ubuntu archive, including libpython2.7-minimal, python2.7-minimal, python-minimal, python2.7, libpython2.7-stdlib, python, fail2ban, python3-pyinotify, and whois. The total size of the downloaded packages is 4370 kB. The terminal then shows the progress of unpacking the packages, including libpython2.7-minimal, python2.7-minimal, python-minimal, and libpython2.7-stdlib. The installation is complete.

```
vagrant@ubuntu:~$ echo "Juan Navarro Maldonado | practica8 | Ubuntu 18.04" > info.txt
vagrant@ubuntu:~$ clear
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | practica8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo apt-get install fail2ban
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libpython-stdlib libpython2.7-minimal libpython2.7-stdlib python
  python-minimal python2.7 python2.7-minimal python3-pyinotify whois
Suggested packages:
  mailx monit sqlite3 python-doc python-tk python2.7-doc binutils
  binfmt-support python-pyinotify-doc
The following NEW packages will be installed:
  fail2ban libpython-stdlib libpython2.7-minimal libpython2.7-stdlib python
  python-minimal python2.7 python2.7-minimal python3-pyinotify whois
0 upgraded, 10 newly installed, 0 to remove and 7 not upgraded.
Need to get 4370 kB of archives.
After this operation, 18.9 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libpython2.7-minimal amd64 2.7.17-1~18.04ubuntu1.11 [335 kB]
Get:2 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 python2.7-minimal amd64 2.7.17-1~18.04ubuntu1.11 [1294 kB]
Get:3 http://archive.ubuntu.com/ubuntu bionic/main amd64 python-minimal amd64 2.7.15~rc1-1 [28.1 kB]
Get:4 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libpython2.7-stdlib amd64 2.7.17-1~18.04ubuntu1.11 [1919 kB]
Get:5 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 python2.7 amd64 2.7.17-1~18.04ubuntu1.11 [248 kB]
Get:6 http://archive.ubuntu.com/ubuntu bionic/main amd64 libpython-stdlib amd64 2.7.15~rc1-1 [7620 B]
Get:7 http://archive.ubuntu.com/ubuntu bionic/main amd64 python amd64 2.7.15~rc1-1 [140 kB]
Get:8 http://archive.ubuntu.com/ubuntu bionic/universe amd64 fail2ban all 0.10.2-2 [329 kB]
Get:9 http://archive.ubuntu.com/ubuntu bionic/main amd64 python3-pyinotify all 0.9.6-1 [24.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu bionic/main amd64 whois amd64 5.3.0 [43.7 kB]
Fetched 4370 kB in 1s (8081 kB/s)
Selecting previously unselected package libpython2.7-minimal:amd64.
(Reading database ... 60336 files and directories currently installed.)
Preparing to unpack .../0-libpython2.7-minimal_2.7.17-1~18.04ubuntu1.11_amd64.deb ...
Unpacking libpython2.7-minimal:amd64 (2.7.17-1~18.04ubuntu1.11) ...
Selecting previously unselected package python2.7-minimal.
Preparing to unpack .../1-python2.7-minimal_2.7.17-1~18.04ubuntu1.11_amd64.deb ...
Unpacking python2.7-minimal (2.7.17-1~18.04ubuntu1.11) ...
Selecting previously unselected package python-minimal.
Preparing to unpack .../2-python-minimal_2.7.15~rc1-1_amd64.deb ...
Unpacking python-minimal (2.7.15~rc1-1) ...
Selecting previously unselected package libpython2.7-stdlib:amd64.
Preparing to unpack .../3-libpython2.7-stdlib_2.7.17-1~18.04ubuntu1.11_amd64.deb ...
Unpacking libpython2.7-stdlib:amd64 (2.7.17-1~18.04ubuntu1.11) ...
```


Debemos de asegurarnos antes de que la configuración sea correcta, para ello nos metemos en el archivos de configuración que hemos copiado en /etc/fail2ban/jail.local y añadimos en el apartado de sshd enabled = true, también podemos ver en dicho fichero los parámetros por defecto, por ejemplo el bantime por defecto es de 10 minutos así como el máximo de intentos es 5, en mi caso voy a bajarlo a 2 intentos para realizar las pruebas posteriormente.

```
vagrant@ubuntu: ~  
vagrant@ubuntu:~$ cat info.txt  
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04  
vagrant@ubuntu:~$ cat info.txt  
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04  
vagrant@ubuntu:~$ clear  
vagrant@ubuntu:~$ cat info.txt  
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04  
vagrant@ubuntu:~$ cat /etc/fail2ban/jail.local | grep --color "enabled"  
# enabled = true  
# "enabled" enables the jails.  
# true: jail will be enabled and log files will get monitored for changes  
# false: jail is not enabled  
enabled = true  
vagrant@ubuntu:~$ cat /etc/fail2ban/jail.local | grep --color "bantime"  
# For example to change the default bantime for all jails and to enable the  
# bantime = 1h  
# "bantime" is the number of seconds that a host is banned.  
bantime = 10m  
action_ = %(banaction)s[name=__name__s, bantime="%(bantime)s", port="%(port)s", protocol="%(protocol)s", chain="%(chain)s"]  
action_mw = %(banaction)s[name=__name__s, bantime="%(bantime)s", port="%(port)s", protocol="%(protocol)s", chain="%(chain)s"]  
action_mwl = %(banaction)s[name=__name__s, bantime="%(bantime)s", port="%(port)s", protocol="%(protocol)s", chain="%(chain)s"]  
action_xarf = %(banaction)s[name=__name__s, bantime="%(bantime)s", port="%(port)s", protocol="%(protocol)s", chain="%(chain)s"]  
bantime = 48h  
bantime = 1w  
# consider low maxretry and a long bantime  
bantime = 1h  
vagrant@ubuntu:~$
```

Ahora comprobamos el estado del cliente:

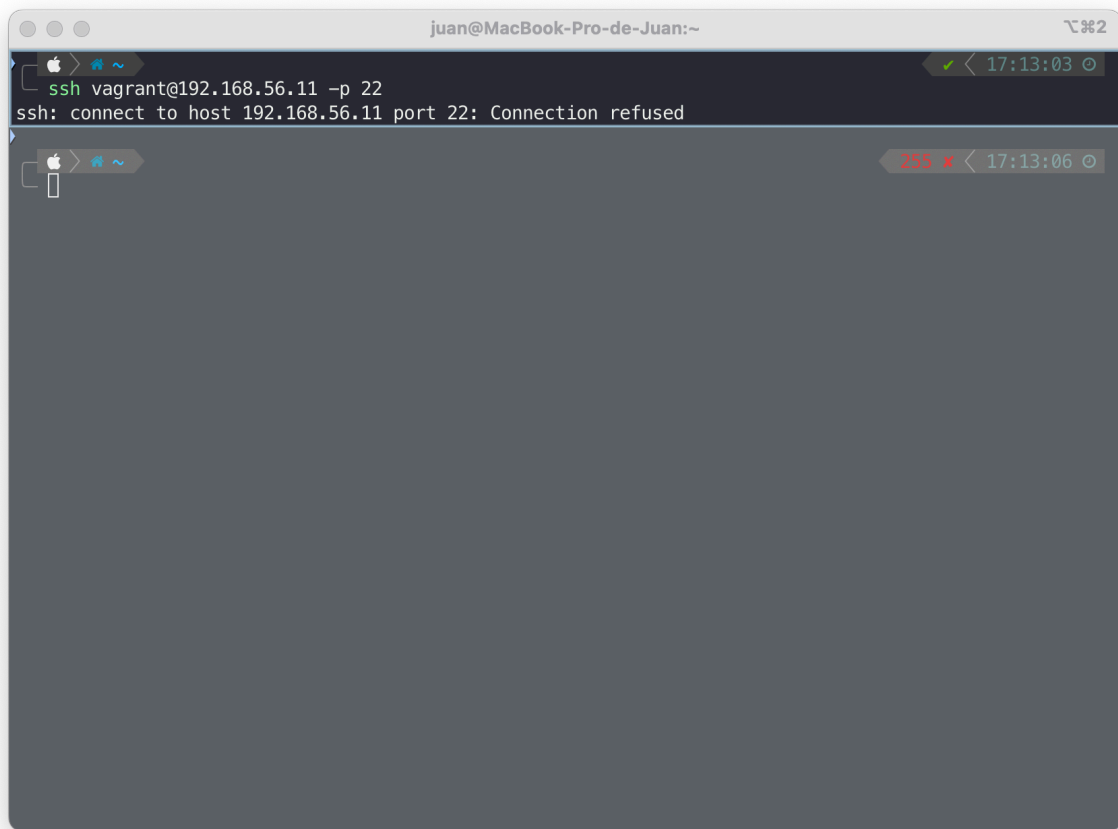
A terminal window titled 'vagrant@ubuntu: ~' with standard Ubuntu window controls (red, yellow, green buttons). The terminal shows the following commands and output:

```
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | practica8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
`- Jail list:  sshd
vagrant@ubuntu:~$
```

Por defecto podemos ver como se activa la jaula de ssh

Comprobación del acceso SSH

Para ello vamos a simular intentos fallidos desde mi máquina anfitriona, como vemos una vez que realizamos los intentos fallidos nos va a dar el siguiente problema:



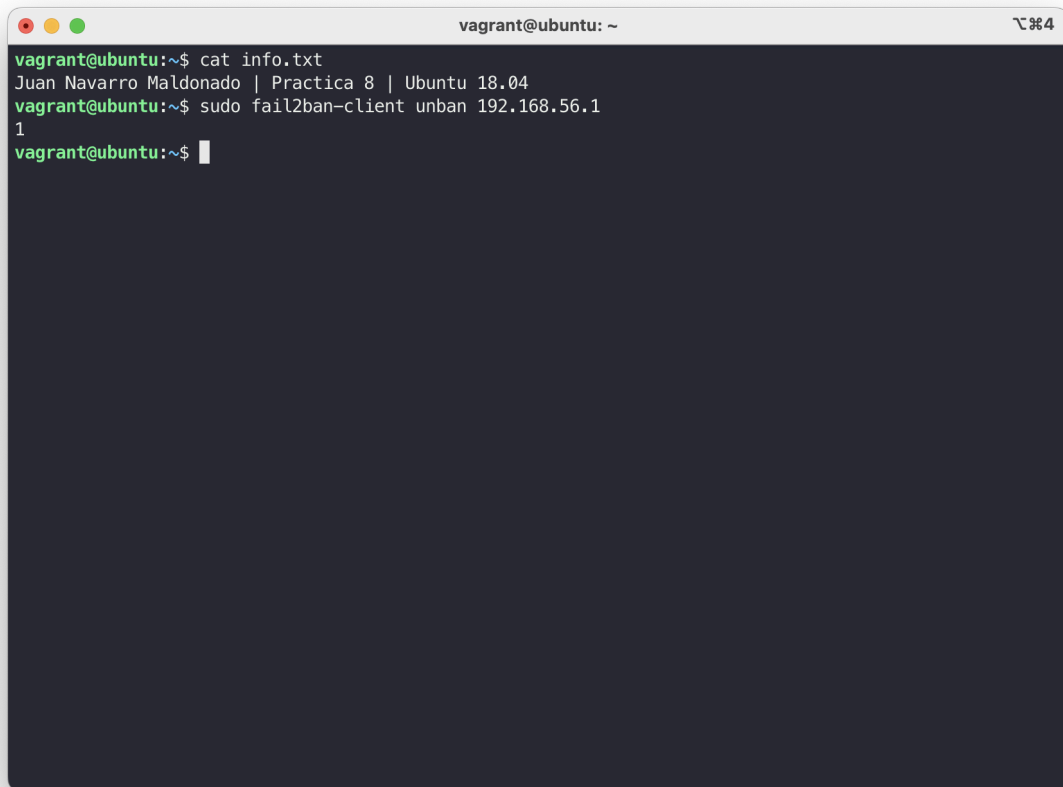
```
juan@MacBook-Pro-de-Juan:~  
ssh vagrant@192.168.56.11 -p 22  
ssh: connect to host 192.168.56.11 port 22: Connection refused
```

The screenshot shows a macOS terminal window with a dark background. The title bar at the top reads 'juan@MacBook-Pro-de-Juan:~'. The terminal content shows a successful execution of the command 'ssh vagrant@192.168.56.11 -p 22', followed by the error message 'ssh: connect to host 192.168.56.11 port 22: Connection refused'. The window's status bar at the bottom right shows a red 'x' icon, indicating an error, and the time '17:13:06'.

Y podemos ver en el status fail2ban para verificarlo:

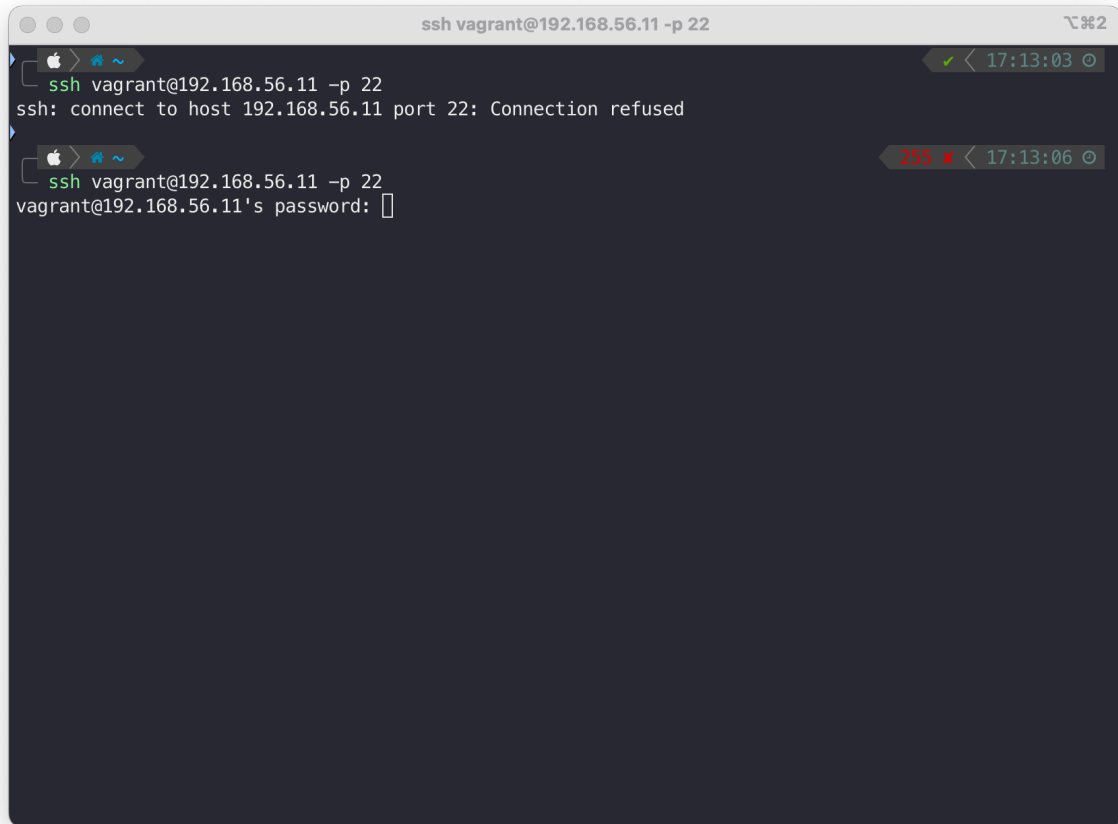
```
vagrant@ubuntu: ~  
vagrant@ubuntu:~$ cat info.txt  
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04  
vagrant@ubuntu:~$ sudo fail2ban-client status  
Status  
|- Number of jail:      1  
  '- Jail list:        sshd  
vagrant@ubuntu:~$ sudo fail2ban-  
fail2ban-client      fail2ban-python      fail2ban-regex      fail2ban-server      fail2ban-testcases  
vagrant@ubuntu:~$ sudo fail2ban-client status sshd  
Status for the jail: sshd  
|- Filter  
|  |- Currently failed: 1  
|  |- Total failed:     10  
|  '- File list:        /var/log/auth.log  
- Actions  
  |- Currently banned: 1  
  |- Total banned:     1  
  '- Banned IP list:    192.168.56.1  
vagrant@ubuntu:~$
```

Si queremos desbanear una dirección, con el siguiente comando:

A terminal window titled 'vagrant@ubuntu: ~' with a dark background. It shows the execution of two commands. The first command is 'cat info.txt', which outputs 'Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04'. The second command is 'sudo fail2ban-client unban 192.168.56.1', which outputs '1'. The prompt 'vagrant@ubuntu:~\$' is visible at the end of the second line.

```
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo fail2ban-client unban 192.168.56.1
1
vagrant@ubuntu:~$
```

Y podemos comprobar que se ha realizado correctamente probando a establecer una conexión por ssh:



```
ssh vagrant@192.168.56.11 -p 22
ssh vagrant@192.168.56.11 -p 22
ssh: connect to host 192.168.56.11 port 22: Connection refused

ssh vagrant@192.168.56.11 -p 22
vagrant@192.168.56.11's password: 
```

The image shows a macOS terminal window titled "ssh vagrant@192.168.56.11 -p 22". It displays two failed SSH connection attempts. The first attempt shows the command "ssh vagrant@192.168.56.11 -p 22" followed by the error message "ssh: connect to host 192.168.56.11 port 22: Connection refused". The second attempt shows the same command, followed by the password prompt "vagrant@192.168.56.11's password:" and a cursor. The terminal has a dark background and a light-colored prompt. The window title bar shows standard macOS window controls and a status bar on the right indicating the connection status and time.

2.- Instalación de Google Authenticator

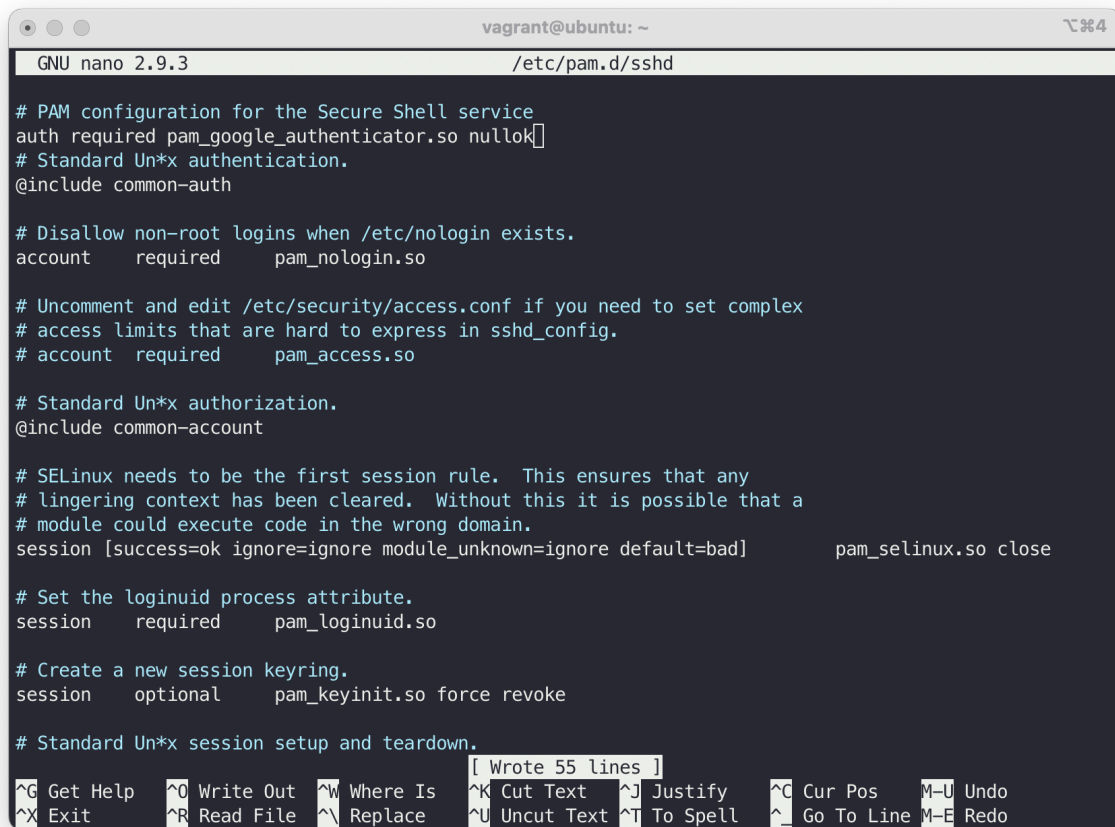
Instalación

Lo primero es instalar el módulo PAM siguiendo las indicaciones del guión con el siguiente comando:

```
vagrant@ubuntu: ~  
vagrant@ubuntu:~$ cat info.txt  
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04  
vagrant@ubuntu:~$ sudo apt-get install libpam-google-authenticator  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following additional packages will be installed:  
  libqrencode3  
The following NEW packages will be installed:  
  libpam-google-authenticator libqrencode3  
0 upgraded, 2 newly installed, 0 to remove and 7 not upgraded.  
Need to get 56.8 kB of archives.  
After this operation, 183 kB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 http://archive.ubuntu.com/ubuntu bionic/universe amd64 libqrencode3 amd64 3.4.4-1build1 [23.9 kB]  
Get:2 http://archive.ubuntu.com/ubuntu bionic/universe amd64 libpam-google-authenticator amd64 20170702-1 [32.9 kB]  
Fetched 56.8 kB in 1s (84.7 kB/s)  
Selecting previously unselected package libqrencode3:amd64.  
(Reading database ... 61578 files and directories currently installed.)  
Preparing to unpack .../libqrencode3_3.4.4-1build1_amd64.deb ...  
Unpacking libqrencode3:amd64 (3.4.4-1build1) ...  
Selecting previously unselected package libpam-google-authenticator.  
Preparing to unpack .../libpam-google-authenticator_20170702-1_amd64.deb ...  
Unpacking libpam-google-authenticator (20170702-1) ...  
Setting up libqrencode3:amd64 (3.4.4-1build1) ...  
Setting up libpam-google-authenticator (20170702-1) ...  
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...  
Processing triggers for libc-bin (2.27-3ubuntu1.6) ...  
vagrant@ubuntu:~$
```

Tenemos que editar el fichero `/etc/pam.d/sshd` y añadimos al comienzo del fichero la siguiente línea:

```
auth required pam_google_authenticator.so nullok
```



```
GNU nano 2.9.3 /etc/pam.d/sshd

# PAM configuration for the Secure Shell service
auth required pam_google_authenticator.so nullok
# Standard Un*x authentication.
@include common-auth

# Disallow non-root logins when /etc/nologin exists.
account required pam_nologin.so

# Uncomment and edit /etc/security/access.conf if you need to set complex
# access limits that are hard to express in sshd_config.
# account required pam_access.so

# Standard Un*x authorization.
@include common-account

# SELinux needs to be the first session rule. This ensures that any
# lingering context has been cleared. Without this it is possible that a
# module could execute code in the wrong domain.
session [success=ok ignore=ignore module_unknown=ignore default=bad] pam_selinux.so close

# Set the loginuid process attribute.
session required pam_loginuid.so

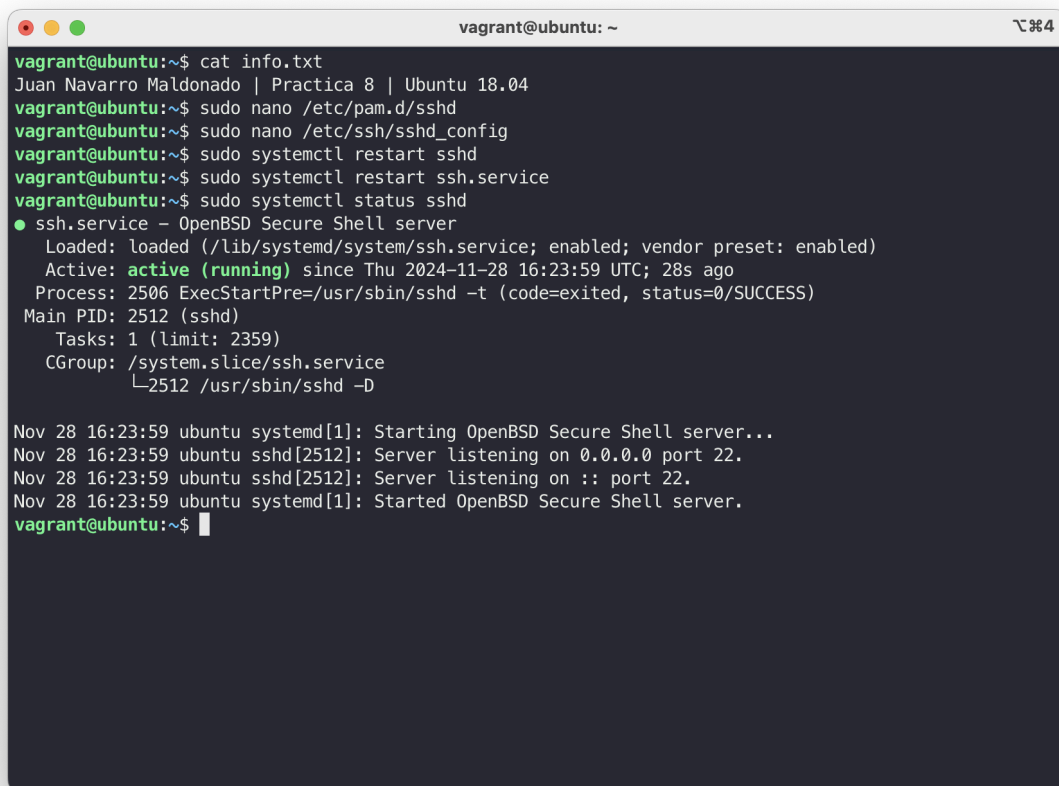
# Create a new session keyring.
session optional pam_keyinit.so force revoke

# Standard Un*x session setup and teardown.

^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text   ^J Justify    ^C Cur Pos    M-U Undo
^X Exit      ^R Read File  ^\ Replace   ^U Uncut Text ^T To Spell   ^_ Go To Line  M-E Redo
```


También debemos de modificar el fichero /etc/ssh/sshd_config para habilitar el challengeResponseAuthentication

Una vez cumplidos estos pasos debemos de reiniciar el servicio de ssh:

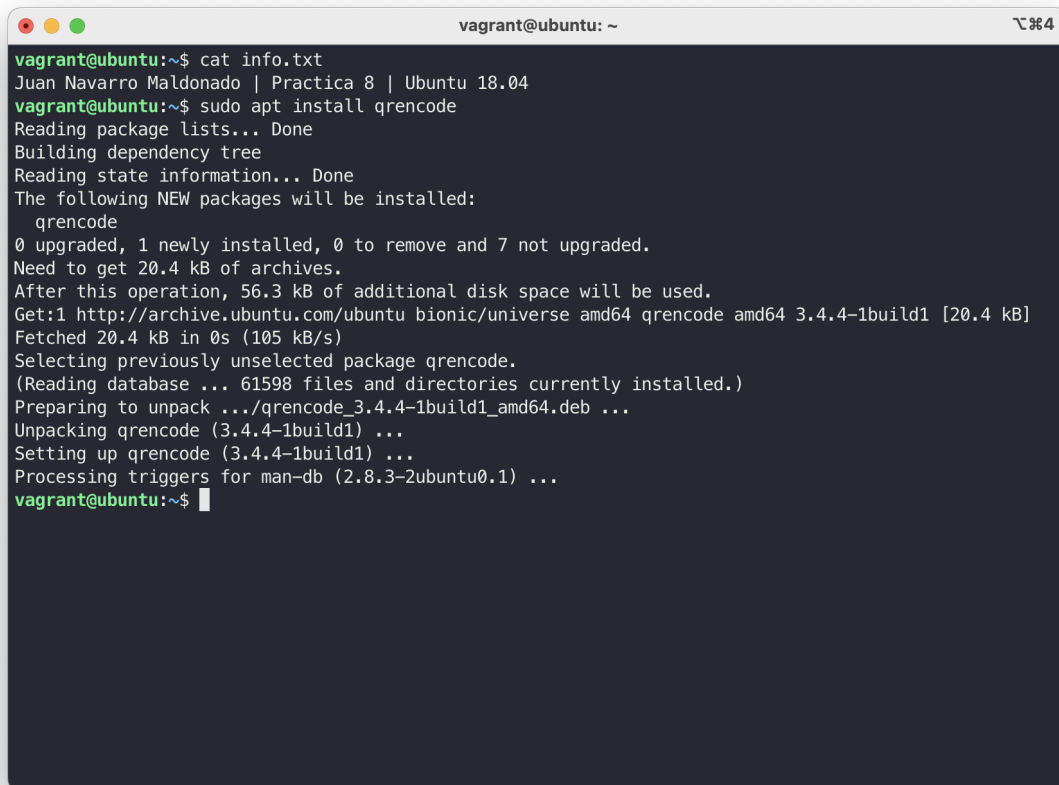
A terminal window titled 'vagrant@ubuntu: ~' with standard window controls. The terminal shows a series of commands and their outputs. First, 'cat info.txt' displays a line 'Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04'. Then, 'sudo nano /etc/pam.d/sshd' and 'sudo nano /etc/ssh/sshd_config' are executed. This is followed by 'sudo systemctl restart sshd' and 'sudo systemctl restart ssh.service'. Finally, 'sudo systemctl status sshd' is run, showing the status of the 'ssh.service' as 'active (running)' since 2024-11-28 16:23:59 UTC. Below this, system logs show the service starting and listening on port 22. The prompt returns to 'vagrant@ubuntu:~\$' at the end.

```
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo nano /etc/pam.d/sshd
vagrant@ubuntu:~$ sudo nano /etc/ssh/sshd_config
vagrant@ubuntu:~$ sudo systemctl restart sshd
vagrant@ubuntu:~$ sudo systemctl restart ssh.service
vagrant@ubuntu:~$ sudo systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2024-11-28 16:23:59 UTC; 28s ago
     Process: 2506 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 2512 (sshd)
      Tasks: 1 (limit: 2359)
     CGroup: /system.slice/ssh.service
            └─2512 /usr/sbin/sshd -D

Nov 28 16:23:59 ubuntu systemd[1]: Starting OpenBSD Secure Shell server...
Nov 28 16:23:59 ubuntu sshd[2512]: Server listening on 0.0.0.0 port 22.
Nov 28 16:23:59 ubuntu sshd[2512]: Server listening on :: port 22.
Nov 28 16:23:59 ubuntu systemd[1]: Started OpenBSD Secure Shell server.
vagrant@ubuntu:~$
```

Y con esto pasaremos a la instalación de qrencode, que es una herramienta de línea de comandos que nos va a permitir generar códigos QR a partir de texto, esto junto con google Authenticator, nos va a permitir generar códigos QR para el sistema de verificación de dos factores, teniendo este QR la información del código temporal.

Para ello vamos a instalarlo con el siguiente comando:

A terminal window titled 'vagrant@ubuntu: ~' with a dark background and light text. It shows the execution of 'cat info.txt' and 'sudo apt install qrencode'. The output of the installation command is displayed, including package lists, dependency tree, state information, and the list of new packages to be installed (qrencode). It also shows the disk space requirements and the progress of downloading and installing the package. The terminal ends with the prompt 'vagrant@ubuntu:~\$' and a cursor.

```
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo apt install qrencode
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  qrencode
0 upgraded, 1 newly installed, 0 to remove and 7 not upgraded.
Need to get 20.4 kB of archives.
After this operation, 56.3 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu bionic/universe amd64 qrencode amd64 3.4.4-1build1 [20.4 kB]
Fetched 20.4 kB in 0s (105 kB/s)
Selecting previously unselected package qrencode.
(Reading database ... 61598 files and directories currently installed.)
Preparing to unpack .../qrencode_3.4.4-1build1_amd64.deb ...
Unpacking qrencode (3.4.4-1build1) ...
Setting up qrencode (3.4.4-1build1) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
vagrant@ubuntu:~$
```

Finalmente ejecutaremos google-authenticator y nos dará nuestro código qr con la información, que escanear en el móvil para obtener mi código 2FA:

```
vagrant@ubuntu: ~  
Setting up qrencode (3.4.4-1build1) ...  
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...  
vagrant@ubuntu:~$ google-authenticator  
  
Do you want authentication tokens to be time-based (y/n) y  
Warning: pasting the following URL into your browser exposes the OTP secret to Google:  
https://www.google.com/chart?chs=200x200&chld=M|0&cht=qr&chl=otpauth://totp/vagrant@ubuntu%3Fsecret%  
3DQWP6QHJS70XR6ALUR4QYJ2J3JA%26issuer%3Dubuntu  
  

```

17:30



Google Authenticator

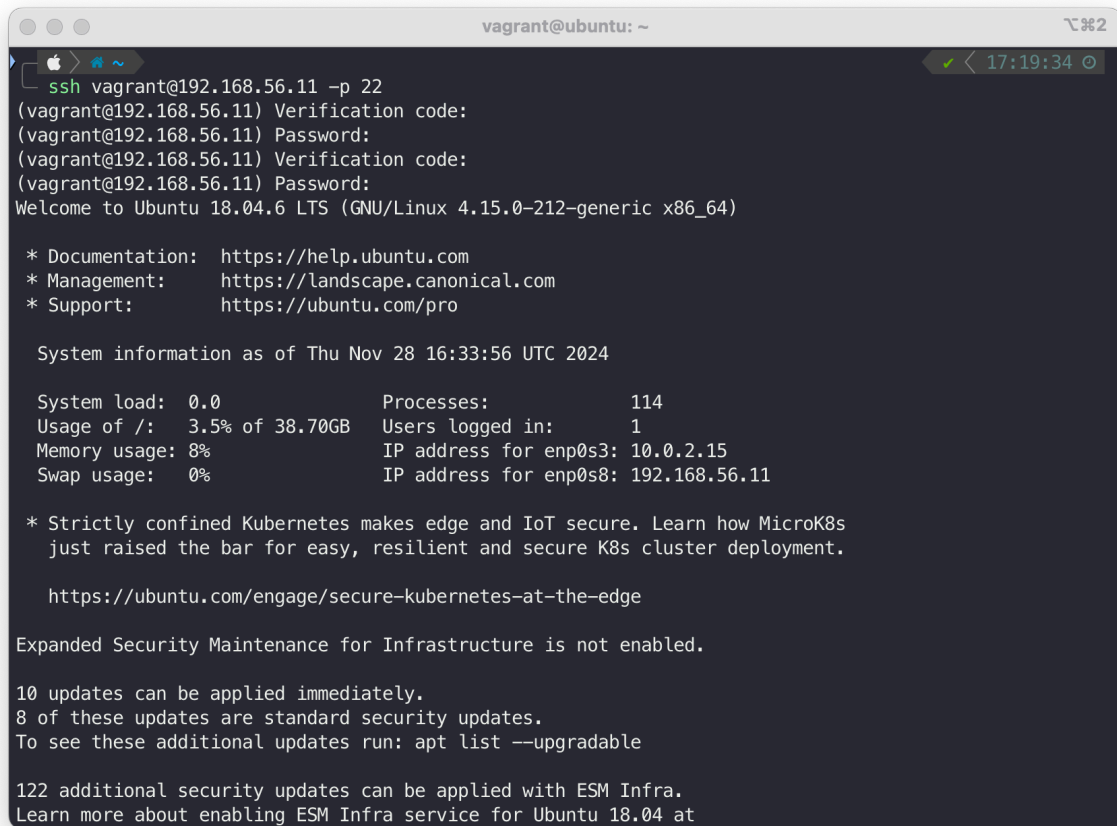


Buscar...

- [Redacted entry] 
- [Redacted entry] 
- [Redacted entry] 
- [Redacted entry] 
- ubuntu: vagrant@ubuntu
690 913 



Por último verificamos que todo se ha hecho correctamente entrando por ssh y vemos que nos pide el código de verificación junto con la contraseña para entrar por ssh



```
vagrant@ubuntu: ~
ssh vagrant@192.168.56.11 -p 22
(vagrant@192.168.56.11) Verification code:
(vagrant@192.168.56.11) Password:
(vagrant@192.168.56.11) Verification code:
(vagrant@192.168.56.11) Password:
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-212-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Thu Nov 28 16:33:56 UTC 2024

System load:  0.0               Processes:            114
Usage of /:   3.5% of 38.70GB   Users logged in:     1
Memory usage: 8%               IP address for enp0s3: 10.0.2.15
Swap usage:   0%               IP address for enp0s8: 192.168.56.11

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

  https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Infrastructure is not enabled.

10 updates can be applied immediately.
8 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

122 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 18.04 at
```

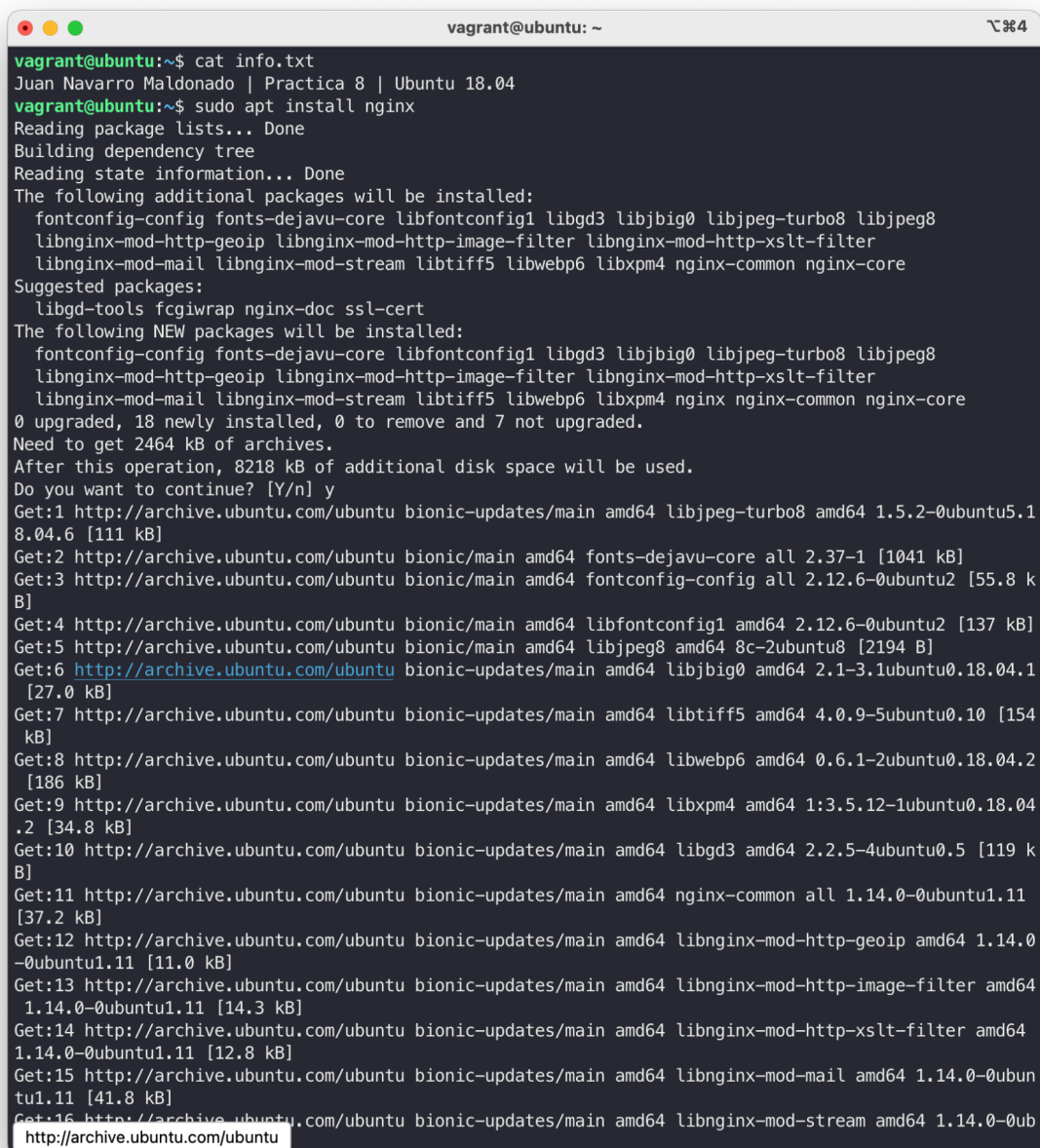
3.- (Opcional) Supervisión de NGINX mediante Fail2ban

Modificación del fichero jail.conf

Como ya he definido en la directiva global en los apartados anteriores en enabled, es decir que ya he activado todas las directivas, no tengo que realizar este paso.

Instalación de NGINX en la máquina Ubuntu

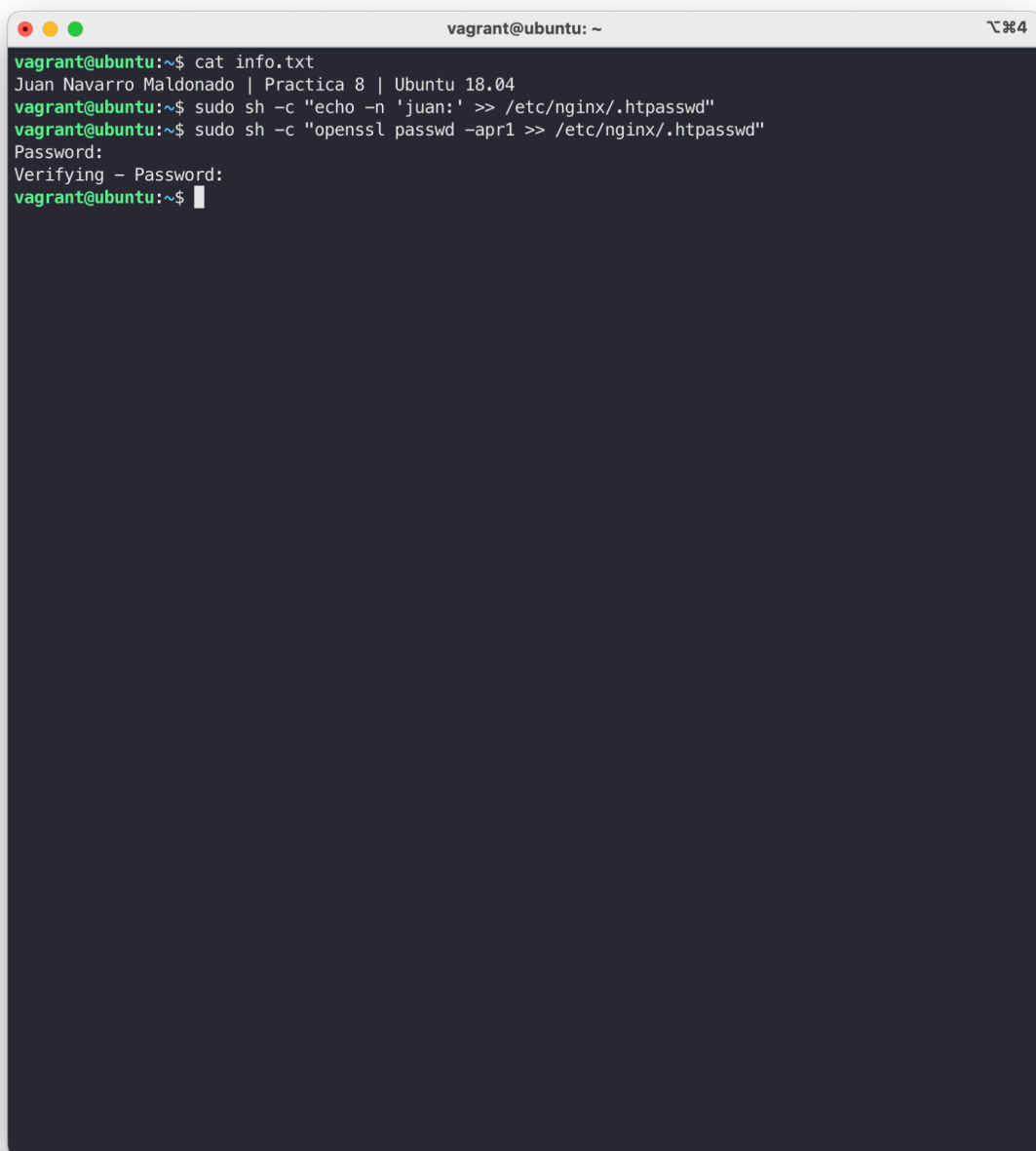
Instalamos nginx con `sudo apt install nginx` en nuestra máquina virtual



```
vagrant@ubuntu: ~  
vagrant@ubuntu:~$ cat info.txt  
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04  
vagrant@ubuntu:~$ sudo apt install nginx  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following additional packages will be installed:  
  fontconfig-config fonts-dejavu-core libfontconfig1 libgd3 libjpeg-turbo8 libjpeg8  
  libnginx-mod-http-geoip libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter  
  libnginx-mod-mail libnginx-mod-stream libtiff5 libwebp6 libxpm4 nginx-common nginx-core  
Suggested packages:  
  libgd-tools fcgiwrap nginx-doc ssl-cert  
The following NEW packages will be installed:  
  fontconfig-config fonts-dejavu-core libfontconfig1 libgd3 libjpeg-turbo8 libjpeg8  
  libnginx-mod-http-geoip libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter  
  libnginx-mod-mail libnginx-mod-stream libtiff5 libwebp6 libxpm4 nginx nginx-common nginx-core  
0 upgraded, 18 newly installed, 0 to remove and 7 not upgraded.  
Need to get 2464 kB of archives.  
After this operation, 8218 kB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libjpeg-turbo8 amd64 1.5.2-0ubuntu5.1  
8.04.6 [111 kB]  
Get:2 http://archive.ubuntu.com/ubuntu bionic/main amd64 fonts-dejavu-core all 2.37-1 [1041 kB]  
Get:3 http://archive.ubuntu.com/ubuntu bionic/main amd64 fontconfig-config all 2.12.6-0ubuntu2 [55.8 k  
B]  
Get:4 http://archive.ubuntu.com/ubuntu bionic/main amd64 libfontconfig1 amd64 2.12.6-0ubuntu2 [137 kB]  
Get:5 http://archive.ubuntu.com/ubuntu bionic/main amd64 libjpeg8 amd64 8c-2ubuntu8 [2194 B]  
Get:6 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libjpeg-turbo8 amd64 1.5.2-0ubuntu5.1  
8.04.6 [111 kB]  
Get:7 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libtiff5 amd64 4.0.9-5ubuntu0.10 [154  
kB]  
Get:8 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libwebp6 amd64 0.6.1-2ubuntu0.18.04.2  
[186 kB]  
Get:9 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libxpm4 amd64 1:3.5.12-1ubuntu0.18.04  
.2 [34.8 kB]  
Get:10 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libgd3 amd64 2.2.5-4ubuntu0.5 [119 k  
B]  
Get:11 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 nginx-common all 1.14.0-0ubuntu1.11  
[37.2 kB]  
Get:12 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libnginx-mod-http-geoip amd64 1.14.0  
-0ubuntu1.11 [11.0 kB]  
Get:13 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libnginx-mod-http-image-filter amd64  
1.14.0-0ubuntu1.11 [14.3 kB]  
Get:14 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libnginx-mod-http-xslt-filter amd64  
1.14.0-0ubuntu1.11 [12.8 kB]  
Get:15 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libnginx-mod-mail amd64 1.14.0-0ubun  
tu1.11 [41.8 kB]  
Get:16 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libnginx-mod-stream amd64 1.14.0-0ub  
tu1.11 [41.8 kB]  
http://archive.ubuntu.com/ubuntu
```

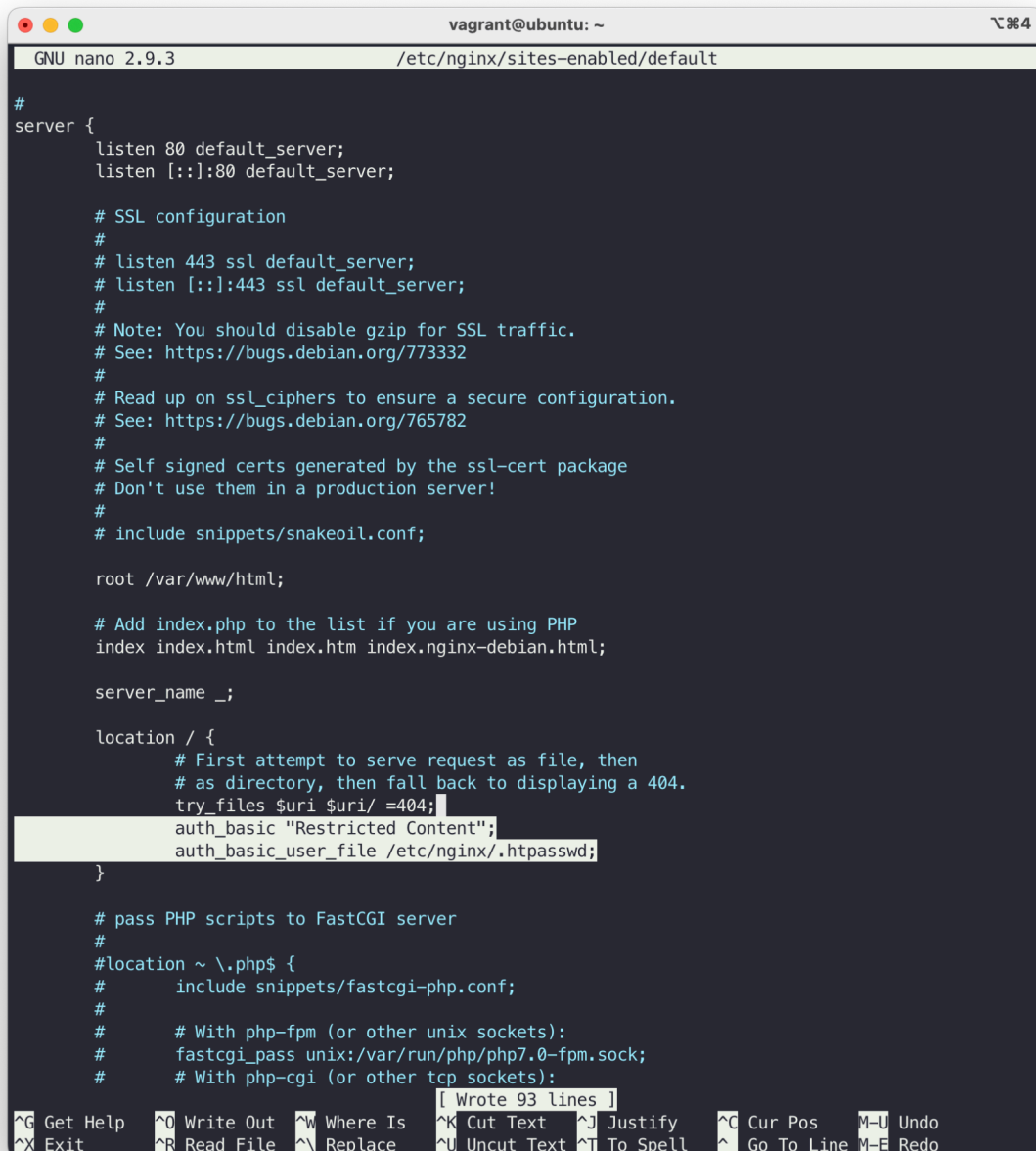
Creación de una página con acceso mediante autenticación

Creamos el fichero /etc/nginx/.htpasswd con las claves:

A terminal window titled 'vagrant@ubuntu: ~' with standard window controls. The terminal shows the following commands and output:

```
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo sh -c "echo -n 'juan:' >> /etc/nginx/.htpasswd"
vagrant@ubuntu:~$ sudo sh -c "openssl passwd -apr1 >> /etc/nginx/.htpasswd"
Password:
Verifying - Password:
vagrant@ubuntu:~$
```

También debemos de modificar el fichero /etc/nginx/sites-enabled/default:



```
GNU nano 2.9.3 /etc/nginx/sites-enabled/default

#
server {
    listen 80 default_server;
    listen [::]:80 default_server;

    # SSL configuration
    #
    # listen 443 ssl default_server;
    # listen [::]:443 ssl default_server;
    #
    # Note: You should disable gzip for SSL traffic.
    # See: https://bugs.debian.org/773332
    #
    # Read up on ssl_ciphers to ensure a secure configuration.
    # See: https://bugs.debian.org/765782
    #
    # Self signed certs generated by the ssl-cert package
    # Don't use them in a production server!
    #
    # include snippets/snakeoil.conf;

    root /var/www/html;

    # Add index.php to the list if you are using PHP
    index index.html index.htm index.nginx-debian.html;

    server_name _;

    location / {
        # First attempt to serve request as file, then
        # as directory, then fall back to displaying a 404.
        try_files $uri $uri/ =404;
        auth_basic "Restricted Content";
        auth_basic_user_file /etc/nginx/.htpasswd;
    }

    # pass PHP scripts to FastCGI server
    #
    #location ~ \.php$ {
    #    include snippets/fastcgi-php.conf;
    #
    #    # With php-fpm (or other unix sockets):
    #    fastcgi_pass unix:/var/run/php/php7.0-fpm.sock;
    #    # With php-cgi (or other tcp sockets):
    #    fastcgi_pass 127.0.0.1:9000;
    #}

    # Wrote 93 lines
}

^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos   M-U Undo
^X Exit      ^R Read File  ^\ Replace   ^U Uncut Text ^T To Spell  ^_ Go To Line M-E Redo
```

y por ultimo reiniciamos el servicio

También pondremos un código básico para testear las credenciales con html

Comprobación entre nodos el bloqueo en los accesos a las páginas

Si entramos a la página podemos ver que nos pide los credenciales para acceder

The image shows a web browser window with a login dialog box. The browser's address bar displays 'Iniciar sesión en localhost'. The dialog box is titled 'Iniciar sesión en localhost:8080' and includes a warning: 'Tu contraseña se enviará sin encriptar.' Below this, there are two input fields: 'Nombre de usuario' and 'Contraseña'. A checkbox labeled 'Guardar esta contraseña' is positioned below the password field. At the bottom of the dialog, there are two buttons: 'Cancelar' and 'Inicia sesión'.

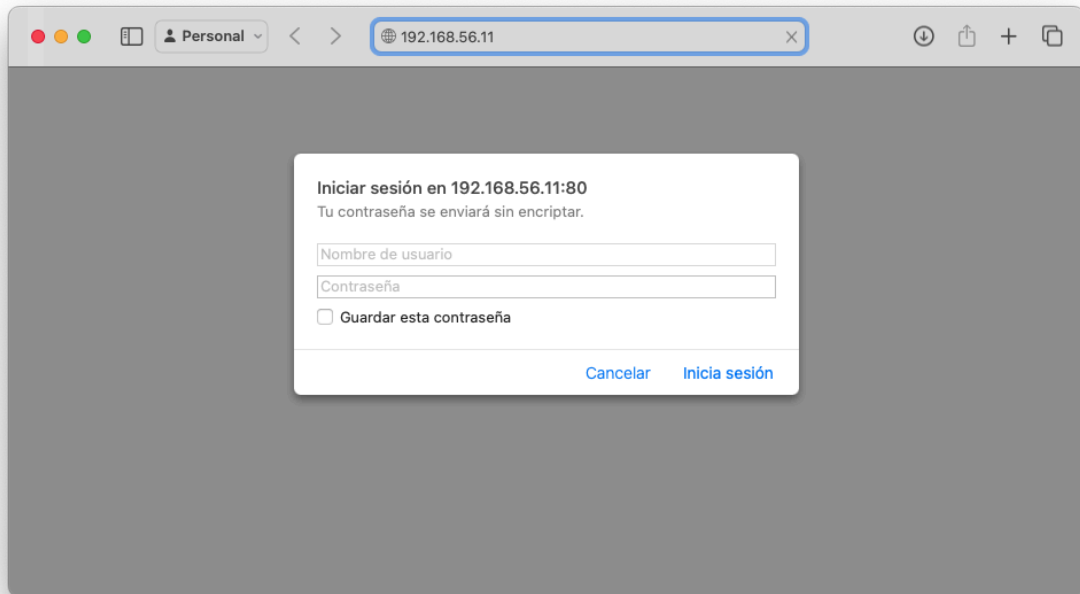
Iniciar sesión en localhost:8080
Tu contraseña se enviará sin encriptar.

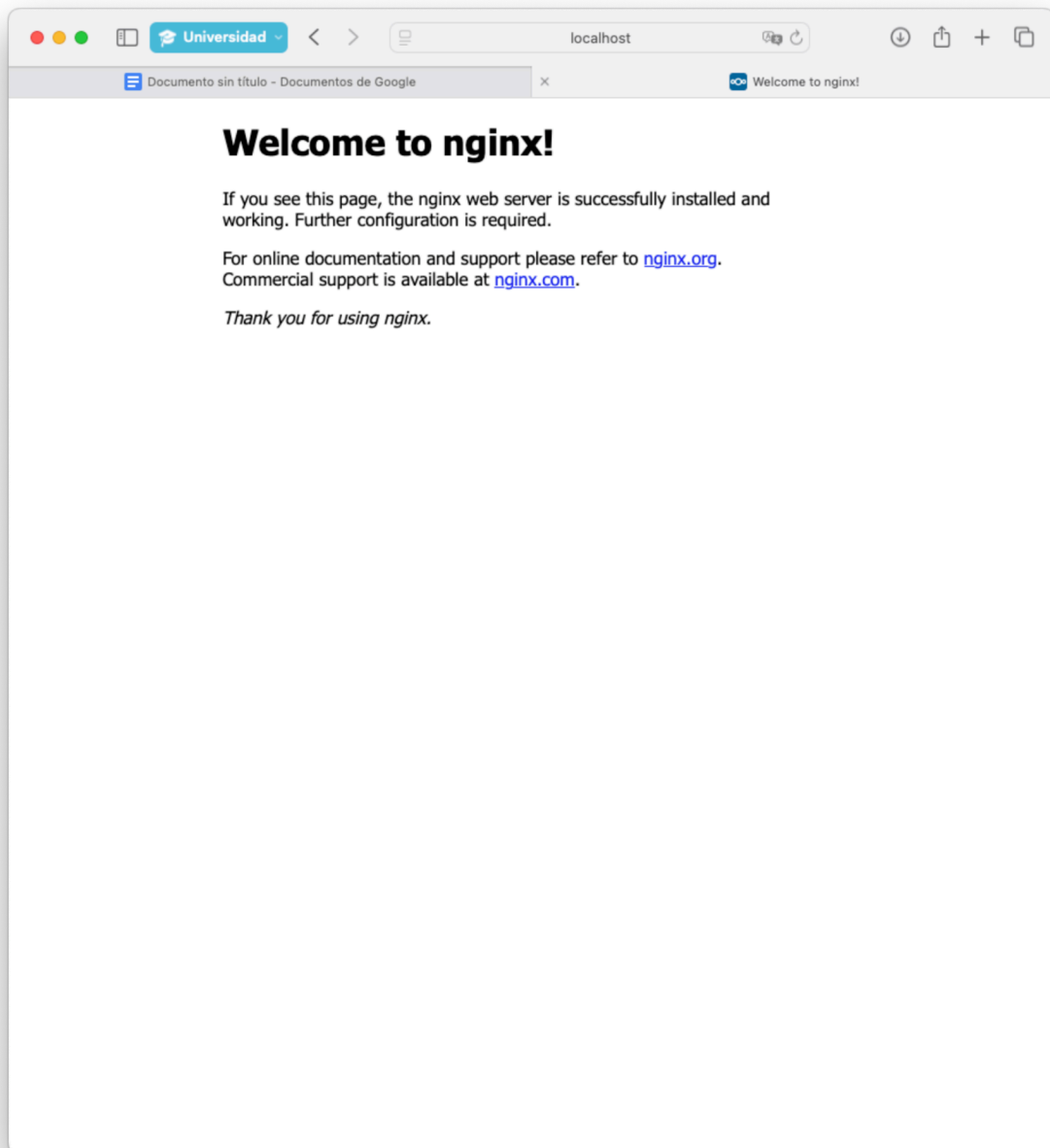
Nombre de usuario
Contraseña

☐ Guardar esta contraseña

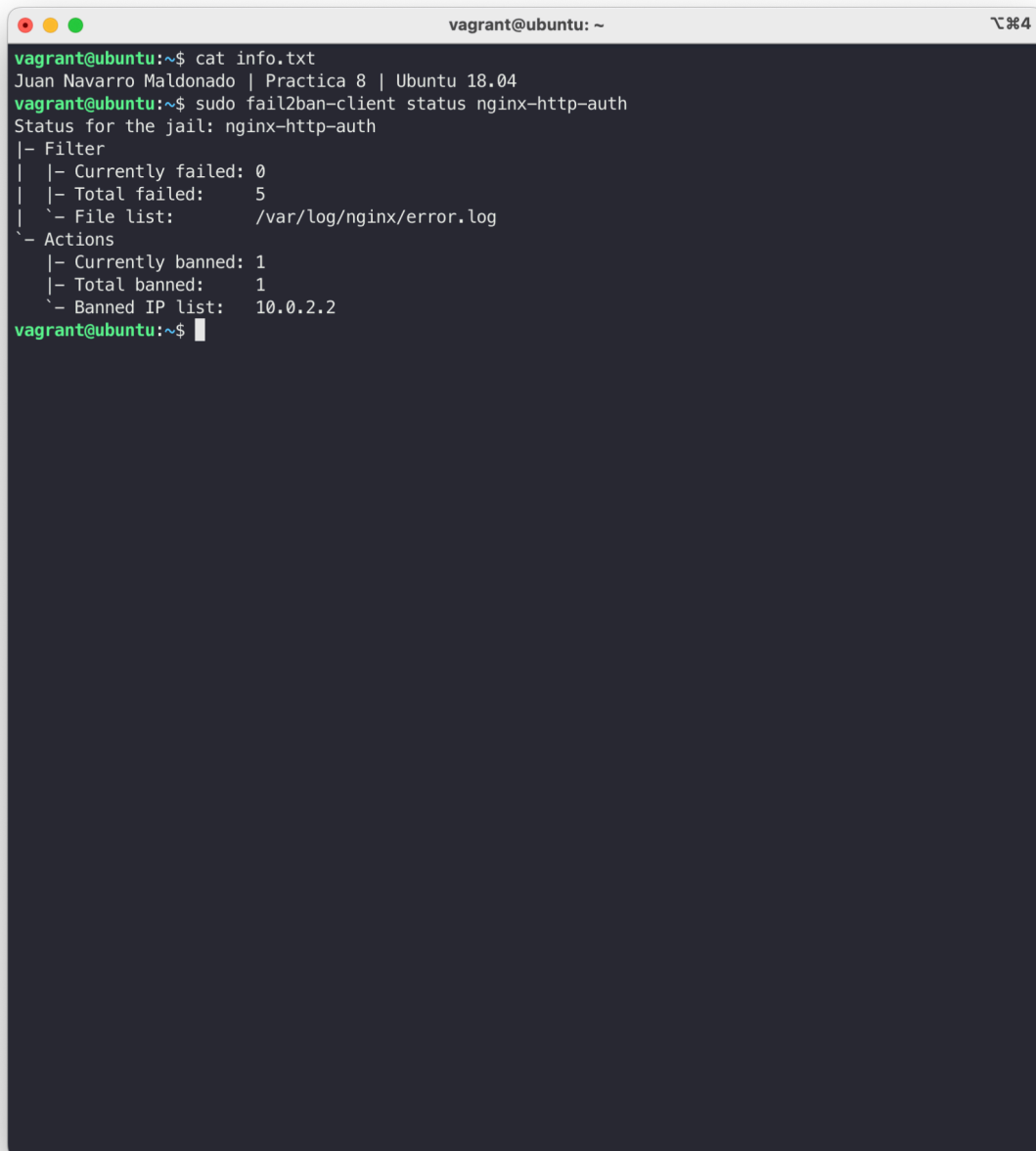
Cancelar Inicia sesión

Nota-> He entrado por localhost ya que en el vagrantfile tengo los puertos mapeados del 80 al 8080 aunque tambien se podria entrar con la dirección de nuestra máquina ubuntu como se muestra en la siguiente imagen



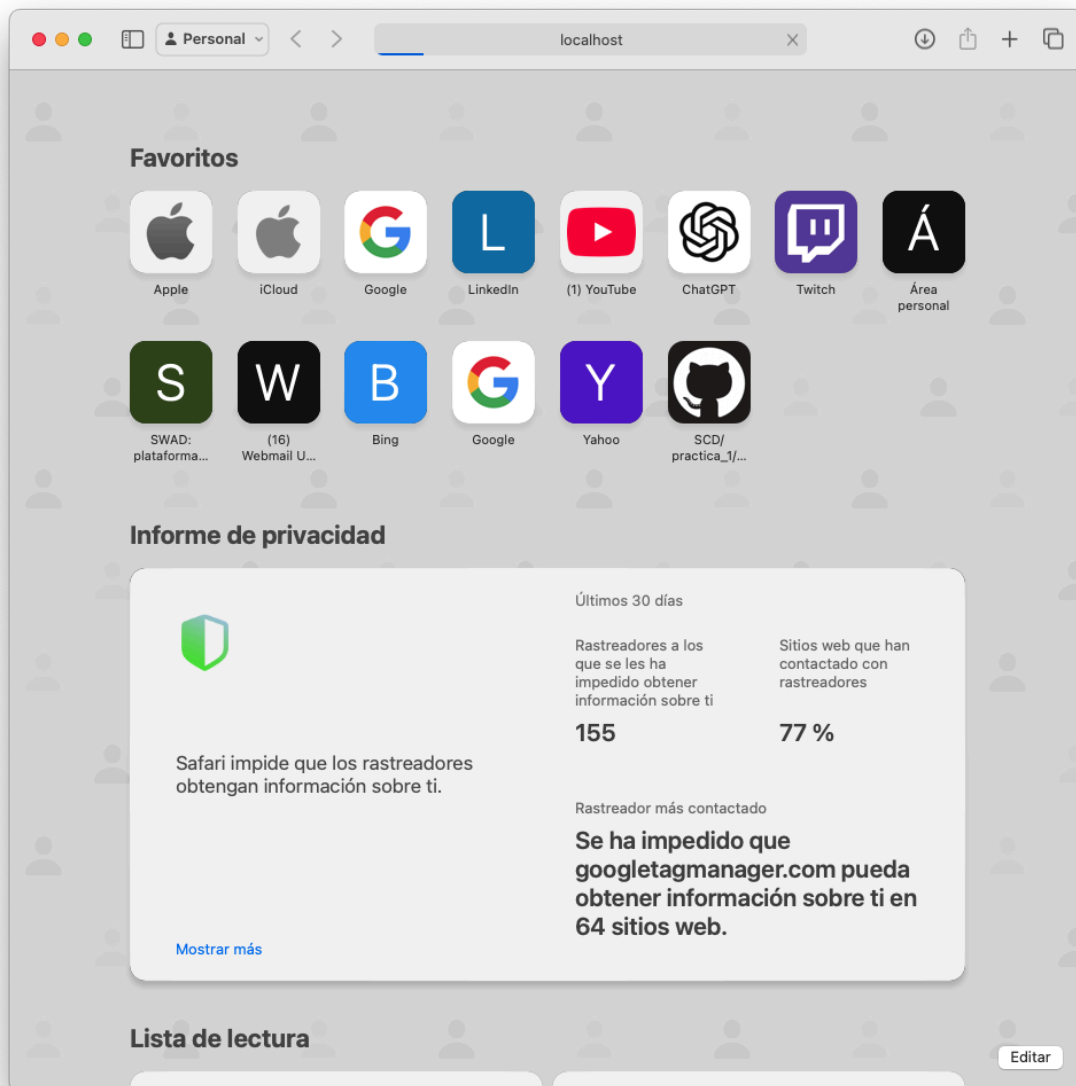


También podemos verificar los fallos ingresando incorrectamente las credenciales



```
vagrant@ubuntu: ~  
vagrant@ubuntu:~$ cat info.txt  
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04  
vagrant@ubuntu:~$ sudo fail2ban-client status nginx-http-auth  
Status for the jail: nginx-http-auth  
|- Filter  
| |- Currently failed: 0  
| |- Total failed: 5  
| `-- File list: /var/log/nginx/error.log  
`- Actions  
   |- Currently banned: 1  
   |- Total banned: 1  
   `-- Banned IP list: 10.0.2.2  
vagrant@ubuntu:~$
```

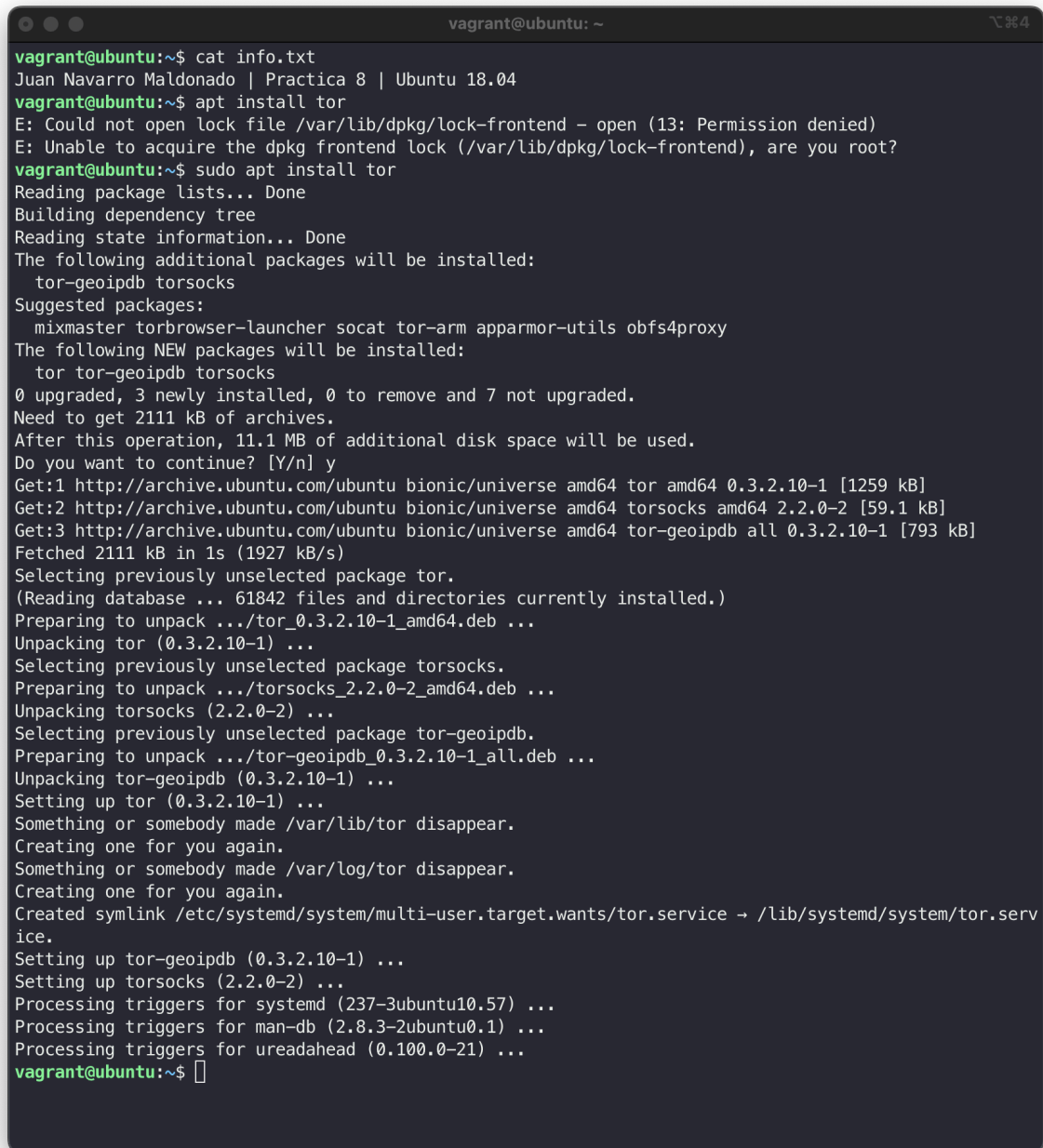
La página no cargará si intentamos acceder



4.- (Opcional) Utilizar la red TOR para acceder directamente a la máquina virtual

Instalación TOR

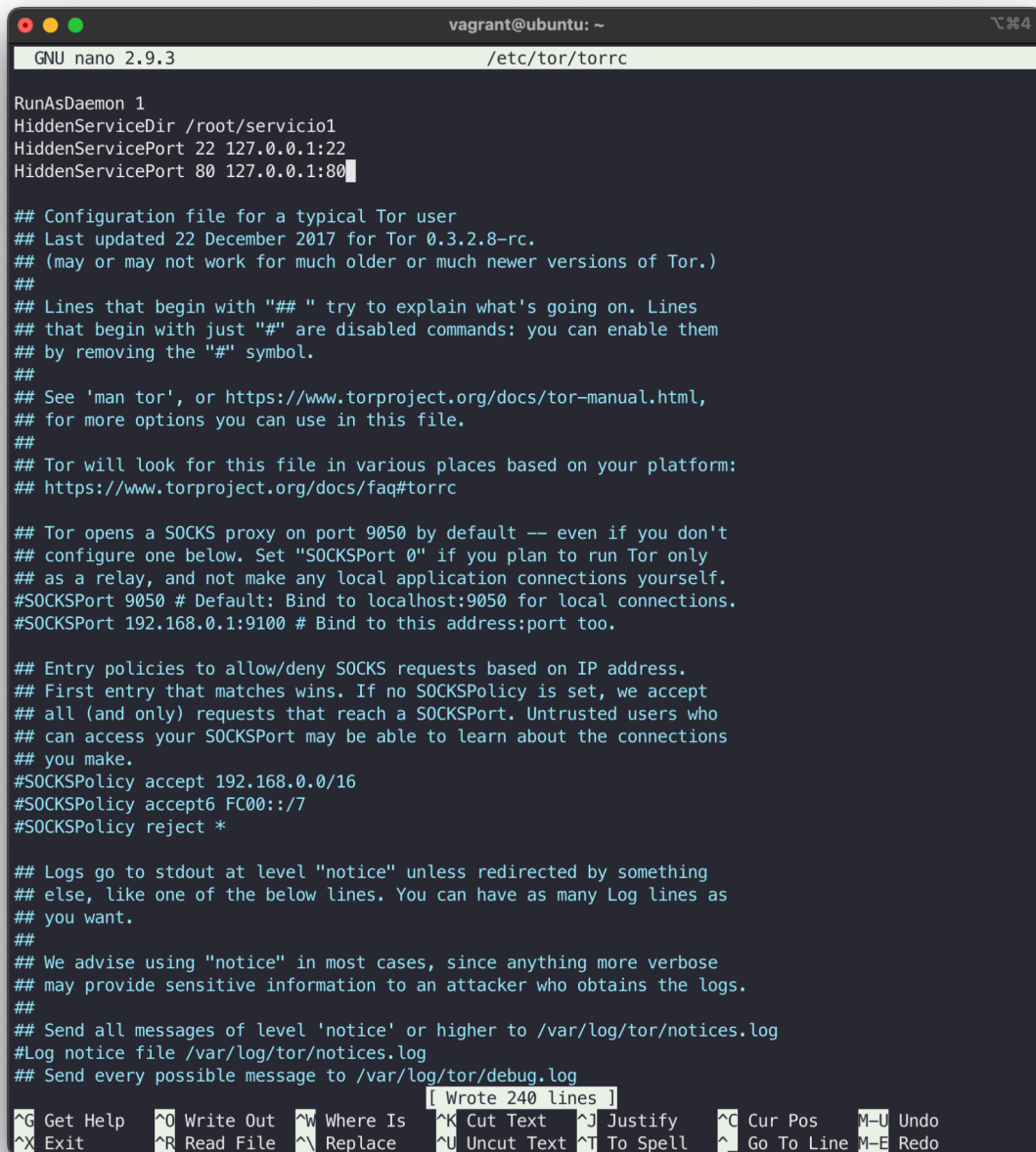
Instalamos tor con el comando `sudo apt install tor`

A terminal window titled 'vagrant@ubuntu: ~' showing the installation of TOR. The user runs 'cat info.txt' which shows 'Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04'. Then they run 'apt install tor' which fails with a permission error. They then run 'sudo apt install tor' which succeeds. The terminal output shows the following additional packages will be installed: tor-geoipdb, torsocks. Suggested packages include mixmaster, torbrowser-launcher, socat, tor-arm, apparmor-utils, and obfs4proxy. The following NEW packages will be installed: tor, tor-geoipdb, and torsocks. The operation will upgrade 0 packages, install 3 new ones, remove 0, and not upgrade 7. It requires 2111 kB of archives and 11.1 MB of additional disk space. The user confirms to continue with 'y'. The terminal shows the download of tor (0.3.2.10-1), torsocks (2.2.0-2), and tor-geoipdb (0.3.2.10-1). It then shows the unpacking and setting up of these packages, including creating symlinks for /var/lib/tor and /var/log/tor, and setting up the systemd service for tor. The installation completes successfully.

```
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ apt install tor
E: Could not open lock file /var/lib/dpkg/lock-frontent - open (13: Permission denied)
E: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontent), are you root?
vagrant@ubuntu:~$ sudo apt install tor
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  tor-geoipdb torsocks
Suggested packages:
  mixmaster torbrowser-launcher socat tor-arm apparmor-utils obfs4proxy
The following NEW packages will be installed:
  tor tor-geoipdb torsocks
0 upgraded, 3 newly installed, 0 to remove and 7 not upgraded.
Need to get 2111 kB of archives.
After this operation, 11.1 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu bionic/universe amd64 tor amd64 0.3.2.10-1 [1259 kB]
Get:2 http://archive.ubuntu.com/ubuntu bionic/universe amd64 torsocks amd64 2.2.0-2 [59.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu bionic/universe amd64 tor-geoipdb all 0.3.2.10-1 [793 kB]
Fetched 2111 kB in 1s (1927 kB/s)
Selecting previously unselected package tor.
(Reading database ... 61842 files and directories currently installed.)
Preparing to unpack .../tor_0.3.2.10-1_amd64.deb ...
Unpacking tor (0.3.2.10-1) ...
Selecting previously unselected package torsocks.
Preparing to unpack .../torsocks_2.2.0-2_amd64.deb ...
Unpacking torsocks (2.2.0-2) ...
Selecting previously unselected package tor-geoipdb.
Preparing to unpack .../tor-geoipdb_0.3.2.10-1_all.deb ...
Unpacking tor-geoipdb (0.3.2.10-1) ...
Setting up tor (0.3.2.10-1) ...
Something or somebody made /var/lib/tor disappear.
Creating one for you again.
Something or somebody made /var/log/tor disappear.
Creating one for you again.
Created symlink /etc/systemd/system/multi-user.target.wants/tor.service → /lib/systemd/system/tor.serv
ice.
Setting up tor-geoipdb (0.3.2.10-1) ...
Setting up torsocks (2.2.0-2) ...
Processing triggers for systemd (237-3ubuntu10.57) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
Processing triggers for ureadahead (0.100.0-21) ...
vagrant@ubuntu:~$
```

Configuración del cliente

Para configurar el cliente debemos de editar el fichero `/etc/tor/torrc` y debemos de añadir las siguientes líneas al fichero:



```
GNU nano 2.9.3 /etc/tor/torrc

RunAsDaemon 1
HiddenServiceDir /root/servicio1
HiddenServicePort 22 127.0.0.1:22
HiddenServicePort 80 127.0.0.1:80

## Configuration file for a typical Tor user
## Last updated 22 December 2017 for Tor 0.3.2.8-rc.
## (may or may not work for much older or much newer versions of Tor.)
##
## Lines that begin with "## " try to explain what's going on. Lines
## that begin with just "##" are disabled commands: you can enable them
## by removing the "##" symbol.
##
## See 'man tor', or https://www.torproject.org/docs/tor-manual.html,
## for more options you can use in this file.
##
## Tor will look for this file in various places based on your platform:
## https://www.torproject.org/docs/faq#torrc

## Tor opens a SOCKS proxy on port 9050 by default -- even if you don't
## configure one below. Set "SOCKSPort 0" if you plan to run Tor only
## as a relay, and not make any local application connections yourself.
#SOCKSPort 9050 # Default: Bind to localhost:9050 for local connections.
#SOCKSPort 192.168.0.1:9100 # Bind to this address:port too.

## Entry policies to allow/deny SOCKS requests based on IP address.
## First entry that matches wins. If no SOCKSPolicy is set, we accept
## all (and only) requests that reach a SOCKSPort. Untrusted users who
## can access your SOCKSPort may be able to learn about the connections
## you make.
#SOCKSPolicy accept 192.168.0.0/16
#SOCKSPolicy accept6 FC00::/7
#SOCKSPolicy reject *

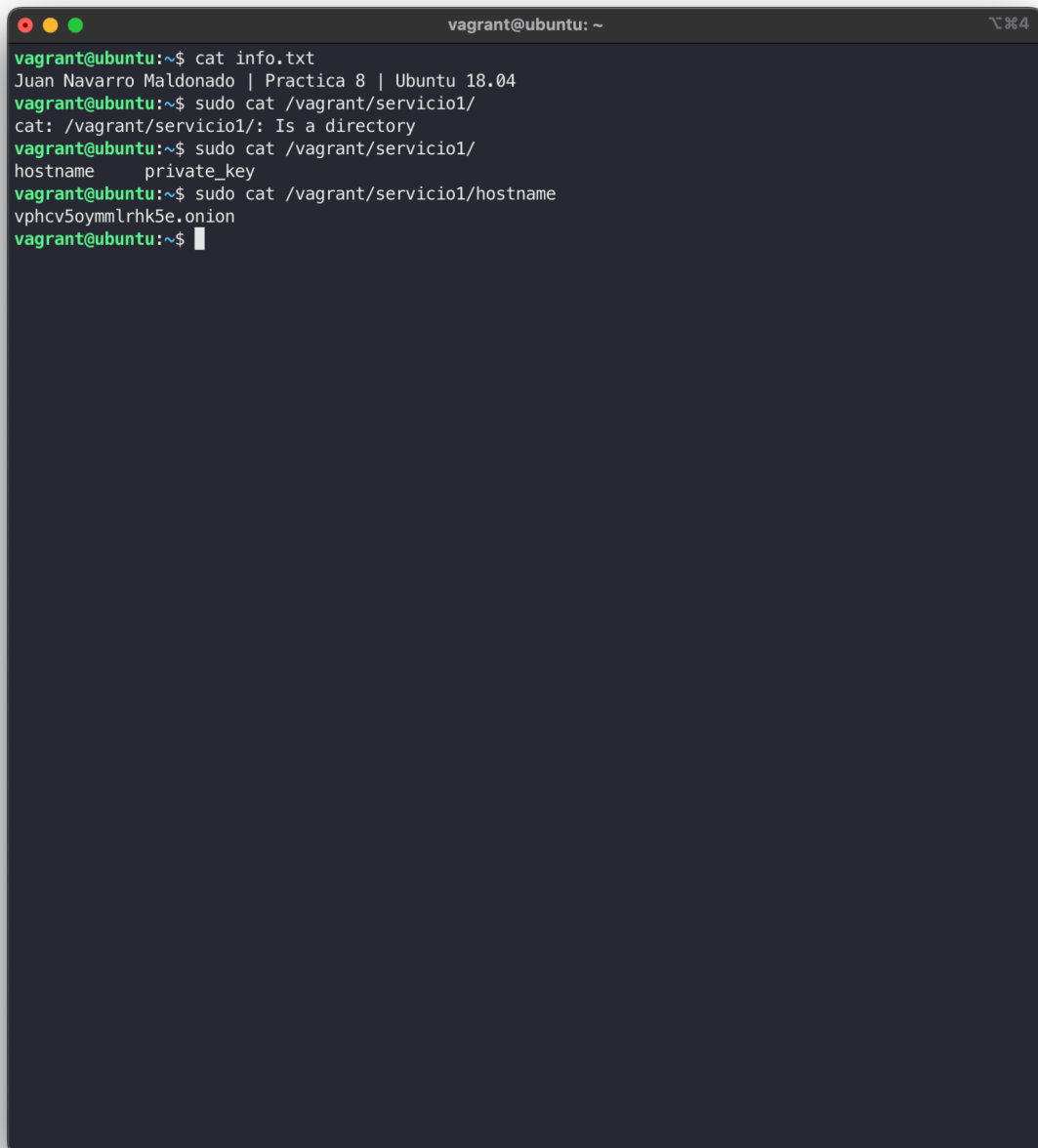
## Logs go to stdout at level "notice" unless redirected by something
## else, like one of the below lines. You can have as many Log lines as
## you want.
##
## We advise using "notice" in most cases, since anything more verbose
## may provide sensitive information to an attacker who obtains the logs.
##
## Send all messages of level 'notice' or higher to /var/log/tor/notices.log
#Log notice file /var/log/tor/notices.log
## Send every possible message to /var/log/tor/debug.log
[ Wrote 240 lines ]

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos M-U Undo
^X Exit ^R Read File ^\ Replace ^U Uncut Text ^T To Spell ^_ Go To Line M-E Redo
```

Por último debemos de iniciar tor habiendo dado previamente los permisos necesario al directorio servicio1

```
vagrant@ubuntu: ~  
vagrant@ubuntu:~$ cat info.txt  
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04  
vagrant@ubuntu:~$ sudo nano /etc/tor/torrc  
vagrant@ubuntu:~$ sudo nano /etc/tor/torrc  
vagrant@ubuntu:~$ tor  
Nov 28 18:00:17.837 [notice] Tor 0.3.2.10 (git-0edaa32732ec8930) running on Linux with Libevent 2.1.8-stable, OpenSSL 1.1.1, Zlib 1.2.11, Liblzma 5.2.2, and Libzstd 1.3.3.  
Nov 28 18:00:17.837 [notice] Tor can't help you if you use it wrong! Learn how to be safe at https://www.torproject.org/download/download#warning  
Nov 28 18:00:17.837 [notice] Read configuration file "/etc/tor/torrc".  
Nov 28 18:00:17.843 [warn] Directory /root/servicio1 cannot be read: Permission denied  
Nov 28 18:00:17.843 [warn] Failed to parse/validate config: Failed to configure rendezvous options. See logs for details.  
Nov 28 18:00:17.843 [err] Reading config failed--see warnings above.  
vagrant@ubuntu:~$ sudo tor  
Nov 28 18:01:08.694 [notice] Tor 0.3.2.10 (git-0edaa32732ec8930) running on Linux with Libevent 2.1.8-stable, OpenSSL 1.1.1, Zlib 1.2.11, Liblzma 5.2.2, and Libzstd 1.3.3.  
Nov 28 18:01:08.694 [notice] Tor can't help you if you use it wrong! Learn how to be safe at https://www.torproject.org/download/download#warning  
Nov 28 18:01:08.694 [notice] Read configuration file "/etc/tor/torrc".  
Nov 28 18:01:08.701 [notice] Scheduler type KIST has been enabled.  
Nov 28 18:01:08.701 [notice] Opening Socks listener on 127.0.0.1:9050  
Nov 28 18:01:08.701 [warn] Could not bind to 127.0.0.1:9050: Address already in use. Is Tor already running?  
Nov 28 18:01:08.702 [warn] Failed to parse/validate config: Failed to bind one of the listener ports.  
Nov 28 18:01:08.702 [err] Reading config failed--see warnings above.  
vagrant@ubuntu:~$ sudo mkdir /root/servicio1  
vagrant@ubuntu:~$ sudo chmod go-rwx /root/servicio1  
vagrant@ubuntu:~$ ls -l /root/  
ls: cannot open directory '/root/': Permission denied  
vagrant@ubuntu:~$ sudo ls -l /root/  
total 4  
drwx----- 2 root root 4096 Nov 28 18:03 servicio1  
vagrant@ubuntu:~$ sudo systemctl start tor  
vagrant@ubuntu:~$ sudo systemctl enable tor  
Synchronizing state of tor.service with SysV service script with /lib/systemd/systemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install enable tor  
vagrant@ubuntu:~$ sudo systemctl status tor  
● tor.service - Anonymizing overlay network for TCP (multi-instance-master)  
   Loaded: loaded (/lib/systemd/system/tor.service; enabled; vendor preset: enabled)  
   Active: active (exited) since Thu 2024-11-28 17:56:54 UTC; 7min ago  
 Main PID: 3881 (code=exited, status=0/SUCCESS)  
    Tasks: 0 (limit: 2359)  
   CGroup: /system.slice/tor.service  
  
Nov 28 17:56:54 ubuntu systemd[1]: Starting Anonymizing overlay network for TCP (multi-instance-master)  
Nov 28 17:56:54 ubuntu systemd[1]: Started Anonymizing overlay network for TCP (multi-instance-master)  
lines 1-9/9 (END)
```


Podemos ver los servicios que nos ha instalado tor ahora

A terminal window titled 'vagrant@ubuntu: ~' with standard Ubuntu window controls. The terminal shows the following commands and output:

```
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo cat /vagrant/servicio1/
cat: /vagrant/servicio1/: Is a directory
vagrant@ubuntu:~$ sudo cat /vagrant/servicio1/
hostname      private_key
vagrant@ubuntu:~$ sudo cat /vagrant/servicio1/hostname
vphcv5oymm1rhk5e.onion
vagrant@ubuntu:~$
```

Con estos pasos ya estaría configurado tor en nuestra máquina virtual.